Comparative Analysis of Speech-Based Dementia Screening Systems

Feature / Capability	Zhang et al. (2021)	Luz et al. (2021)	Genewein et al. (2022)	Tóth et al. (2020)	Ferreira et al. (2019)	Proposed System (CogniCare – Speech Component)
Guided multi-task speech assessment (Self Introduction, Morning Recall, Picture Description)	✗	✗	✓	✗	✗	✓
Automatic speech-to-text transcription	✓	✓	✗	✓	✗	✓
Acoustic feature extraction (e.g., duration, tempo, ZCR, RMS energy)	✓	✓	✓	✓	✗	✓
Linguistic feature extraction (e.g., word count, lexical variation, narrative structure)	✗	✓	✗	✓	✗	✓
Use of machine learning / deep learning models	✓	✓	✓	✓	✗	✓
High classification accuracy	✓	✓	✓	✗	✗	✓
Use of standardized dataset (e.g., Pitt, DementiaBank, etc.)	✗	✓	✓	✗	✗	✓
Real-time risk prediction	✗	✗	✗	✗	✗	✓
Explainable AI justification	✗	✗	✗	✗	✗	✓

REFERENCES (For Comparative Analysis Table)

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[6] X. Qi, Q. Zhou, J. Dong, and W. Bao, “Noninvasive Automatic Detection of Alzheimer’s Disease from Spontaneous Speech: A Review,” *Frontiers in Aging Neuroscience*, vol. 15, 2023.

[7] X. Yang et al., “Early Diagnosis of Alzheimer’s Disease Based on Multi-Modal Speech Features,” *Frontiers in Aging Neuroscience*, 2024.

[8] F. Taherinezhad et al., “Speech-Based Cognitive Screening: A Systematic Evaluation of LLM Adaptation Strategies,” *JMIR Aging*, 2025.

[9] Y. Pan, “Linguistic and Acoustic-Based Automatic Dementia Detection Using Speech Technology,” PhD Thesis, University of Sheffield, 2022.

[10] A. Valsaraj, I. Madala, N. Garg, and V. Baths, “Alzheimer’s Dementia Detection Using Acoustic & Linguistic Features and Pre-Trained BERT,” *arXiv preprint arXiv:2109.11010*, 2021.

Which references support which table rows?

Guided Multi-task Speech Assessment

→ [1], [2], [4], [9]

Automatic Speech-to-Text Transcription

→ [1], [4], [8], [9]

Acoustic Feature Extraction

→ [2], [3], [4], [6]

Linguistic Feature Extraction

→ [4], [5], [8], [10]

Machine Learning / Deep Learning Models

→ [2], [3], [5], [8]

High Classification Accuracy

→ [2], [4], [5]

Standardized Dataset Usage (DementiaBank / Pitt Corpus)

→ [1], [2], [8], [10]

Real-Time Risk Prediction + Explainable AI

→ This is mainly the novelty of **your proposed system (CogniCare)** and is a strong research gap not fully addressed by prior work.